

Day (53)

* Exception Handling in Python :-

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=> Keywords used Exception Handling.

1) try

2) except

3) else

4) finally

5) raise

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* Types of Error :-

- The purpose of Exception Handling is that "To Build Robust (strong) Applications".
- In realtime, to develop any type of application, we need to choose language.
- By using the language, we can develop, compile and execute various programs. During this process, we get various types of errors. They are classified into 3 types

They are:

- ① Compile Time errors.
- ② Logical Errors.
- ③ Runtime errors.

① Compile time error :-

- Compile time errors are those which are occurring during compilation process (`.py` $\rightarrow$ `.pyc`).
- Compile time errors occur due to programmer not following syntaxes of the language.
- Compile time errors solved by programmers during development time.

② Logical time error :-

- Logical errors are those which are occurring during execution time (`.pyc` $\rightarrow$ `pvm`).
- Logical errors occur due to wrong representation/invalid logics of problem.
- Logical errors always give wrong results.
- Logical errors solved by programmers during development time.

③ Runtime Errors (Implementation Errors)

- Runtime errors are those which are occurring during execution / Runtime time.
- Runtime errors due to wrong or invalid inputs entered by end-users or application user.
- By default runtime errors always generates Technical error messages, which are understandable by programmers but not by end-users.
So, industry recommended to convert Technical error messages into user-friendly error messages by using exception handling.

* Building points for learning exception handling :-

①. when the Application user enters invalid / wrong input then we get runtime errors.
(invalid input - - - -> runtime error)

②. All runtime errors by default generates Technical error messages.
(runtime error - - - -> Technical error messages)

③. Definition of Exception :-

↓
Every runtime error is called exception.
(invalid input - - - -> runtime error - - - -> Exception)

Hence every invalid input gives exception.

④. Definition of Exception Handling :-

↓
The process of converting Technical error messages into user-friendly error message is called exception handling.

⑤ When an exception occurs, internally 3 steps takes place they are:

- ① program execution Abnormally terminated by pvm
- ② pvm comes out-of program execution.
- ③ By default pvm generates Technical error message.

⑥. To perform step ①, step ②, and step ③, internally "pvm" creates an object of appropriate exception class.

⑦. When exception occurs in python then pvm creates an object appropriate exception class.

⑧. Hence every exception in python is considered as object and it is created w.r.t appropriate exception class.

Note:

- When the Application user enters valid input then we get result/output. When the Application user enters invalid input then we get exception.

* Types of Exception in python :-

→ In python programming, we have two types of exceptions

they are: ① pre-defined or Built-in exceptions.

② programmer or user or custom defined exception.

① pre-defined or Built-in exceptions:-

→ These exceptions are developed by python lang developers and available in python software and used by python lang. programmers for dealing with universal problem.

⇒ some of the universal problems are :-

- a) Invalid number format (Value error)
- b) Invalid Attribute Names (Attribute error)
- c) Invalid key (Key error)
- d) module does not exist (Module Not Found error)
- e) Wrong index (Index error) ... etc.

(2). programmer or user or custom defined exceptions:-

→ These exceptions developed by Python Lang. programmer and available in Python project and used by other team members of same project for dealing with common problems in project.

⇒ some of the common problems are :-

- a) attempting to enter invalid pin in ATM Applⁿ.
- b) attempting to withdraw more amount than existing Balance.
- c) attempting to enter invalid username and password... etc